



PSL Recognizer: Empowering Communication through AI

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ABSTRACT

The deaf community in Pakistan faces major communication barriers due to the lack of effective PSL (Pakistani Sign Language) recognition systems. While sign languages like ASL have been widely researched, PSL remains underexplored. To bridge this gap, we developed a deep learning-based PSL recognition pipeline using a dataset of 104 daily-use signs, expanded through extensive video augmentation. Among the three evaluated models—ViViT, ResNet18+LSTM, and a proposed 3DCNN + Residual + BiLSTM hybrid—our hybrid model showed superior generalization. This work presents a scalable PSL recognition system and a diverse dataset, contributing to more inclusive communication for Pakistan’s deaf community.

OBJECTIVES

- To bridge the communication gap between the deaf community and the general population in Pakistan.
- To assemble and augment a robust video dataset of 104 daily-use PSL signs.
- To evaluate and compare multiple deep learning models for PSL video recognition.
- To design a hybrid deep learning architecture that effectively captures spatiotemporal features and sequence patterns.
- To achieve high classification accuracy and generalisation on unseen PSL sign data.

METHODOLOGY

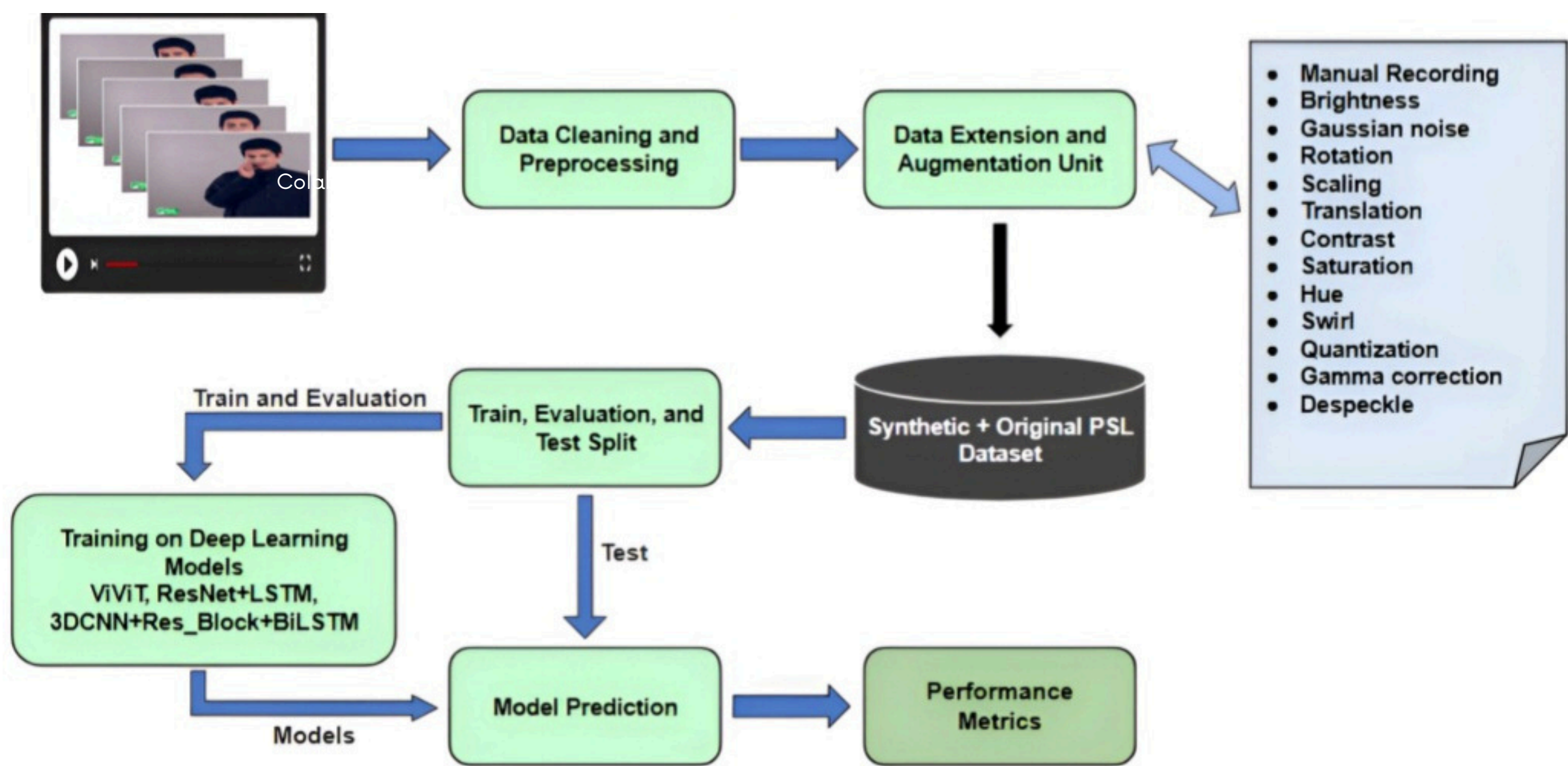


Figure 1: defines the pipeline followed by this study

RESULTS & DISCUSSIONS

We evaluated three deep learning models on our augmented PSL dataset. ViViT achieved 96% training accuracy but overfitted with a 21% performance drop on evaluation. ResNet18+LSTM generalized better but lacked practical performance. Our proposed 3DCNN–Residual–BiLSTM hybrid outperformed both, achieving 83.89% accuracy and strong precision, recall, and F1 scores. These results validate its effectiveness for real-time PSL recognition and its potential to support inclusive communication.

	Train Loss	Eval Loss	Train Acc	Eval Acc	Train Precision	Eval Precision	Train Recall	Eval Recall	Train F1 Score	Eval F1 Score
ResNet+LSTM (70 epochs)	1.07	1.38	0.96	0.84	0.96	0.86	0.96	0.84	0.96	0.84
3DCNN+RES_BLOCK+BiLSTM (10 epochs)	0.47	0.48	0.87	0.86	0.87	0.89	0.87	0.86	0.87	0.86
Video Vision Transformer (ViViT) (25 epochs)	0.25	1.2	0.96	0.75	0.97	0.79	0.96	0.75	0.96	0.75

Figure 2: Evaluation Metrics of the 3 models

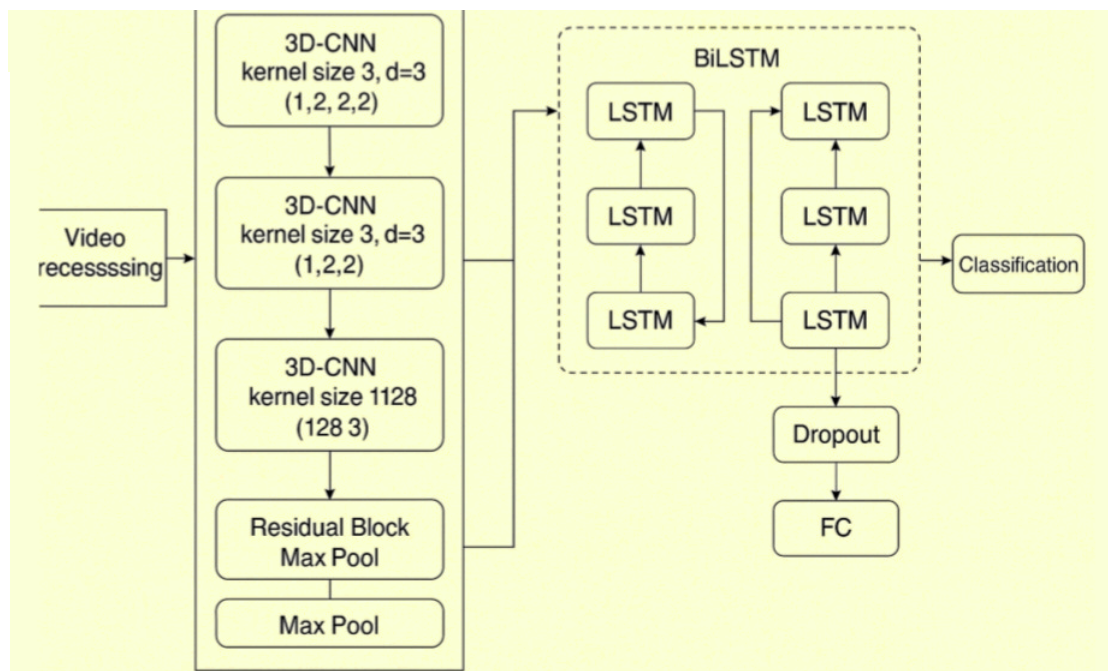


Figure 3: shows architecture incorporated by 3DCNN+ResidualBlock+BiLSTM

CONCLUSION

This work presents a PSL (Pakistani Sign Language) recognition system aimed at bridging the communication gap for the deaf community in Pakistan. A dataset of 104 daily-use signs was augmented at the frame level to boost training diversity. Among three evaluated models, our proposed hybrid architecture—combining 3DCNN, Residual Blocks, and BiLSTM—achieved the best generalization with 83.89% accuracy and an F1 score of 0.84. These results demonstrate the promise of deep learning for real-time PSL translation and inclusive communication.

TECHNOLOGY STACK

